

EC4215 — Business Econometrics and Forecasting

Autumn 2025 Examination · Paper 1 · Full Marking Scheme

Module	EC4215 — Business Econometrics and Forecasting
Internal Examiner	Dr. Turner
Institution	University College Cork
Session	Autumn 2025, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt 2 of 3 (100% of the paper). All questions carry equal weight.
Materials	Non-programmable calculator permitted. Formula sheet and statistical tables (t, F, normal, Durbin-Watson) provided.

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions.

Own-figure rule. This entire paper is calculation-heavy — every t-test, F-test, and Durbin-Watson decision carries forward from an earlier step. An answer that applies the correct method and formula consistently from an earlier arithmetic slip still earns the marks available for the later steps.

Source basis. This scheme follows the module's own course material closely throughout (Studenmund, *Using Econometrics*, and Gujarati, *Econometrics by Example* — the two texts the uploaded course pack draws on), including the exact expected- bias-analysis formula, the Durbin-Watson test's assumptions, and the Breusch-Godfrey test's role when a lagged dependent variable is present.

Section A · Question 1

Interpreting a flight-demand regression, and testing a cinema-attendance regression for significance — 100 marks

A1(a)

[60 marks]

A researcher examining the demand for flights from Ireland to Spain produces: $F_t = 150.8 + 5.2GDP_t + 11.2R_{t-1} - 10.4P_t$, with standard errors (2.4), (4.2), (6.8), $\bar{R}^2=0.67$, $n=20$. (i) Carefully interpret the above findings. (ii) Does the above equation fit your expectations? Why or why not? (iii) If you could add one more variable into the above regression, what would it be and why?

A1(b)

[40 marks]

A researcher examines weekly cinema attendance: $A_t = 996.1 - 4.78P_t + 1.47S_t + 3.45H_t$, with standard errors (2.22), (0.84), (1.14), $\bar{R}^2=0.478$, $n=52$, $RSS=49,016$, $ESS=45,443$. (i) Looking at this output (without carrying out any statistical tests), would you expect the regression to be significant? Explain your answer. (ii) Formally test this regression for significance at the 5% level of significance. Explain the steps you take.

FULL MODEL ANSWER

(i) Interpreting each coefficient: with standard errors given in brackets beneath GDP, $R(t-1)$, and P respectively, each coefficient is interpreted *ceteris paribus* — holding the other explanatory variables constant.

Variable	Coefficient	t-statistic	Interpretation
Intercept	150.8	—	The (largely non-economic, extrapolated) baseline number of seats booked (150,800) if GDP, lagged rainfall, and price were all zero — not a meaningful real-world scenario, but a necessary part of the fitted line
GDP_t	5.2	2.167	A €1bn rise in Irish GDP is associated with 5,200 more seats booked, holding rainfall and price constant — consistent with air travel being a normal good, more disposable income raising discretionary travel demand
R_{t-1}	11.2	2.667	A 1mm rise in the PREVIOUS year's rainfall is associated with 11,200 more seats booked this year, holding GDP and price constant — consistent with a lagged "escape the weather"

effect, where a wetter previous year increases the following year's appetite for a sunshine holiday

P_t	-10.4	-1.529	A €1 rise in the average adult seat price is associated with 10,400 fewer seats booked, holding GDP and rainfall constant — consistent with a standard downward-sloping demand curve
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Statistical significance (t-tests, $df = n - k - 1 = 20 - 3 - 1 = 16$): comparing each t-statistic to the critical value from the appendix t-table at $df=16$.

Variable	t-statistic	Critical t (5%, two-tail, $df=16$)	Significant?
GDP_t	2.167	2.120	Yes — $ t > t_{crit}$
R_{t-1}	2.667	2.120	Yes — $ t > t_{crit}$
P_t	-1.529	2.120	No at 5% two-tail ($ t < t_{crit}$); significant at 10% two-tail / 5% one-tail, where $t_{crit}=1.746$

$\bar{R}^2 = 0.67$: 67% of the variation in flight bookings is explained by the model, after adjusting for the number of explanatory variables and degrees of freedom used — a reasonably strong fit for this type of demand equation.

Mark Allocation — Part (a)(i) (35% of Q. A1(a))

Tier	Marks	What separates this tier
Full credit	76-100%	All four coefficients (including the intercept) correctly interpreted in ceteris paribus, economically-meaningful terms AND all three t-tests correctly calculated and compared to the correct critical value at the correct df AND $\bar{R}^2$ correctly interpreted.
Good	51-75%	Coefficients correctly interpreted with at least two of the three t-tests correctly carried out.
Partial	26-50%	Coefficients interpreted only in terms of sign/direction without ceteris paribus magnitude language, and/or t-tests not attempted.
Minimal	0-25%	No credible interpretation given.

(ii) Does the equation fit expectations? Largely yes, with one variable requiring a specific justifying story. The GDP coefficient's positive sign and the price coefficient's negative sign both match standard economic expectations without needing any special explanation — air travel is easily understood as a normal good that responds negatively to its own price. The lagged rainfall coefficient's positive sign is *not* an obviously expected result on its face — rainfall is an unusual variable to include in a flight-demand equation at all — but it becomes plausible once the specific behavioural story (worse weather last year increasing the following year's appetite to escape to a sunnier destination) is applied, and it is both statistically significant and sensibly sized. Overall, the equation fits expectations reasonably well, provided the unusual rainfall variable's inclusion is specifically justified by that story rather than left unexplained.

Mark Allocation — Part (a)(ii) (15% of Q. A1(a))

Tier	Marks	What separates this tier
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Full credit	76-100%	Correctly identifies GDP and price as matching standard expectations without further justification needed AND correctly identifies the rainfall variable as requiring its own specific behavioural story to make sense, rather than dismissing or accepting it uncritically.
Good	51-75%	A reasonable overall verdict given with some but not full engagement with the rainfall variable's unusual nature.
Partial	26-50%	States the equation "fits expectations" or "doesn't" without engaging each variable individually.
Minimal	0-25%	No credible answer given.

(iii) An additional variable worth adding: the **price of substitute holiday destinations** (e.g. an average price index for comparable sun-holiday destinations such as Portugal or the Canary Islands) would be a strong candidate. Demand for any specific product depends not just on its own price but on the price of its close substitutes — if Spain becomes relatively more or less expensive compared to competing holiday destinations, this should shift demand for Ireland-Spain flights independently of Spain's own seat price or Irish GDP, and omitting it risks omitted variable bias if substitute prices are correlated with any of the included regressors (see Q. A2(b) for the general mechanics of this kind of bias).

EXAM TECHNIQUE

Other well-justified answers — a marketing/advertising spend variable, an exchange rate variable (EUR/GBP, since UK carriers compete on some Ireland-Spain routes), or a dummy variable for a new budget airline entering the route — are equally creditable provided the economic reasoning for why it belongs in a flight-demand equation is clearly given.

Mark Allocation — Part (a)(iii) (10% of Q. A1(a))

Tier	Marks	What separates this tier
Full credit	76-100%	A specific, economically well-motivated additional variable proposed with clear reasoning for why it should affect flight demand and why its omission is a concern.
Good	51-75%	A reasonable variable proposed with some reasoning given.
Partial	26-50%	A variable named without economic reasoning.
Minimal	0-25%	No credible variable proposed.

(b)(i) Would you expect the regression to be significant, without testing? Yes, likely. Although $\bar{R}^2=0.478$ (a moderate, not spectacular, share of variation explained) doesn't look overwhelming on its own, the sample size here is relatively large ($n=52$) relative to the number of explanatory variables ($k=3$) — giving 48 degrees of freedom. With this many degrees of freedom, even a moderate R^2 typically translates into a statistically significant F-statistic for overall significance, since the F-test rewards explanatory power achieved with a large number of observations relative to the number of parameters estimated.

Mark Allocation — Part (b)(i) (10% of Q. A1(b))

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly predicts significance AND correctly reasons from the large sample size relative to k (large degrees of freedom), not just from the $\bar{R}^2$ value in isolation.

Good	51-75%	Correct prediction with some reasoning given.
Partial	26-50%	A prediction given based only on the $\bar{R}^2$ value without considering sample size/degrees of freedom.
Minimal	0-25%	No credible answer given.

(b)(ii) Formal F-test for overall significance: the null hypothesis for the overall F-test is $H_0: \beta_1=\beta_2=\beta_3=0$ (none of the explanatory variables jointly explain any variation in attendance), against H_a : at least one $\beta_k \neq 0$.

Step	Calculation	Result
TSS = RSS + ESS	49,016 + 45,443	94,459
$F = (ESS/k) / (RSS/(n-k-1))$	$(45,443/3) / (49,016/48)$	14.834
Degrees of freedom	$df_1=k=3, df_2=n-k-1=48$	F(3,48)
Critical F(3,48) at 5%	interpolated between F(3,40)=2.84 and F(3,60)=2.76	≈ 2.80

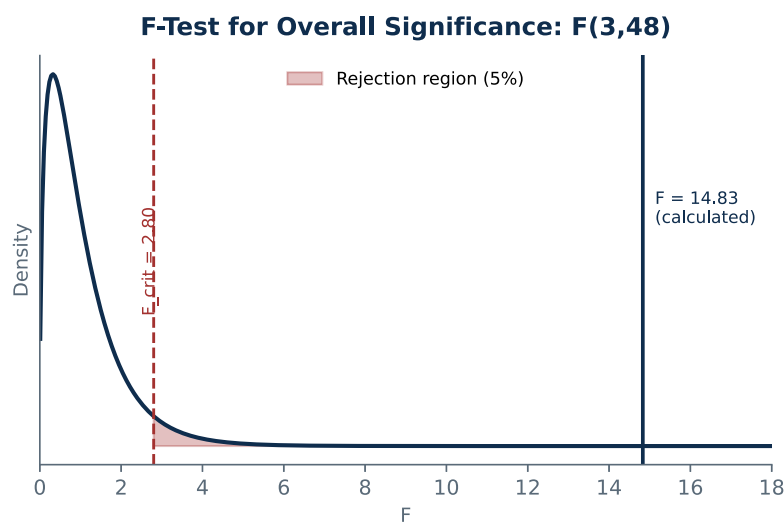


Fig. 1 — The calculated F-statistic (14.83) falls far into the rejection region beyond the critical value (≈ 2.80) — a decisive rejection of the null of no overall significance.

The steps, and the conclusion: (1) state the null and alternative hypotheses; (2) calculate the F-statistic using the ESS and RSS given; (3) determine the correct degrees of freedom ($df_1=k, df_2=n-k-1$); (4) find the critical F-value at the 5% significance level for those degrees of freedom from the F-table; (5) compare. Since the calculated F (14.834) is substantially larger than the critical value (≈ 2.80), we **reject the null hypothesis** — the regression **is statistically significant overall** at the 5% level, confirming the prediction made in Part (b)(i).

Mark Allocation — Part (b)(ii) (30% of Q. A1(b))

Tier	Marks	What separates this tier
Full credit	76-100%	Correct null/alternative hypotheses stated AND F correctly calculated as 14.834 AND correct degrees of freedom identified (3, 48) AND correctly compared to the (interpolated or conservative) critical value AND the correct "reject H_0 , significant" conclusion given.
Good	51-75%	F correctly calculated and degrees of freedom correct, but the critical value comparison or conclusion has an error.
Partial	26-50%	F formula set up correctly but arithmetic error in the final statistic.

Question A1 Total: 100%

Full Marking Scheme · Study-Uni

Q1

EC4215 · Business Econometrics and Forecasting

Autumn 2025 Examination · Paper 1

Section A · Question 2

Comparing two nested electric-car regressions, and expected bias analysis for an omitted variable — 100 marks

A2(a)**[70 marks]**

A researcher examining monthly electric car demand gets: $EC_t = 80.4 - 4.4PC_t - 2.1PE_t + 15.8EM_t$ (Eqn. 2.1), SEs (1.8), (1.3), (5.4), $N=24$, $Adj\ R^2=0.551$. Adding public charging infrastructure gives: $EC_t = 67.1 - 4.5PR_t - 1.9PE_t + 15.7EM_t + 2.1CP_t$ (Eqn. 2.2), SEs (1.7), (0.8), (5.2), (0.8), $N=24$, $Adj\ R^2=0.601$. Should the researcher include CP_t (i.e. go with Eqn. 2.2 instead of Eqn. 2.1)? Explain your answer.

A2(b)**[30 marks]**

A researcher estimates $SALESt = 325.4 + 21.4Pt - 13.1PFt$ for new car sales in Cork city, and is surprised at the positive coefficient on price. She considers two possible omitted variables: $PUBt$ (public transport journeys) and $IN Ct$ (average household income). Use expected bias analysis to determine whether either is a viable candidate as the omitted relevant variable.

FULL MODEL ANSWER

The three pieces of evidence to weigh: deciding whether to add a variable to a regression should draw on more than one diagnostic — the change in explanatory power, the added variable's own statistical significance, and whether the other coefficients remain economically sensible across the two specifications.

1. Adjusted R^2 rises when CP is added: Adjusted R^2 specifically **penalises** adding variables that don't genuinely improve the model's explanatory power — unlike plain R^2 , which mechanically rises whenever any variable is added regardless of its relevance, Adjusted R^2 can fall if a newly added variable isn't pulling its weight. Here, Adjusted R^2 **rises** from 0.551 to 0.601 when CP is added — a genuinely meaningful increase that survives the adjustment penalty, providing strong evidence that CP is a relevant explanatory variable rather than noise.

2. CP is individually statistically significant: testing $H_0: \beta(\text{CP})=0$ with $t = 2.1/0.8 = 2.625$, against a critical $t(0.05, df=24-4-1=19, \text{two-tail}) = 2.093$ — since $2.625 > 2.093$, **CP is statistically significant** at the 5% level, meaning its own coefficient is unlikely to be zero in the population.

3. The other coefficients remain stable and sensible across both specifications: comparing Eqn. 2.1 to Eqn. 2.2, the coefficients on price ($-4.4 \rightarrow -4.5$), electricity price ($-2.1 \rightarrow -1.9$), and employment ($15.8 \rightarrow 15.7$) all stay close in magnitude and keep their expected signs — this stability is a good sign that Eqn. 2.1 wasn't suffering from severe omitted variable bias on those coefficients specifically, but the genuine improvement in fit (points 1 and 2 above) still indicates CP captures real, additional explanatory power that belongs in the model on its own economic merits (more public charging infrastructure plausibly makes EV ownership more practical, directly supporting higher EV demand).

Conclusion: Yes, the researcher should include CP and adopt Eqn. 2.2 — the rise in Adjusted R^2 , CP's own statistical significance, and CP's economically sensible positive sign (more charging infrastructure $\rightarrow$ more EV demand) all point the same direction.

Mark Allocation — Part (a) (70%)

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly identifies the rise in Adjusted R^2 as evidence for inclusion, explaining why Adjusted R^2 (unlike plain R^2) is informative here AND correctly calculates CP's t-statistic (2.625) and compares to the correct critical value (2.093) AND notes the stability of the other coefficients across specifications AND reaches the correct "include CP, adopt Eqn. 2.2" conclusion.
Good	51-75%	Correctly uses the Adjusted R^2 rise and/or CP's significance to recommend inclusion, but doesn't develop both pieces of evidence together.
Partial	26-50%	States a conclusion based only on the Adjusted R^2 change, without testing CP's individual significance.
Minimal	0-25%	No credible comparison of the two equations given.

Setting up the expected bias framework: the expected bias in the (true) coefficient of an included variable X_1 , caused by omitting a genuinely relevant variable X_2 , has the same sign as: [the true coefficient of X_2 on the dependent variable] $\times$ [the sign of the relationship between X_1 and X_2]. A candidate omitted variable is a **viable** explanation for an unexpected result only if the sign of its expected bias matches the direction of the unexpected result actually observed.

The unexpected result to be explained: the observed coefficient on price (+21.4) is positive, where standard demand theory predicts a **negative** price coefficient (higher price $\rightarrow$ lower quantity demanded). We need an omitted variable whose omission would bias the price coefficient **upward (positively)** — turning a truly negative coefficient into the observed positive one.

Candidate 1: PUBt (public transport journeys)

- **True effect of PUB on car sales:** public transport and car ownership are substitute means of getting around — more public transport usage should plausibly mean *fewer* new car sales, so the true coefficient of PUB on SALES is expected to be **negative**.

- **Relationship between P (car price) and PUB:** as new car prices rise, some consumers priced out of the car market would plausibly switch toward public transport instead — suggesting a **positive** relationship between P and PUB (higher car prices associated with more public transport use).
- **Expected bias** = (negative) × (positive) = **negative**.

Conclusion for PUB: the expected bias from omitting PUB is **negative** — the opposite sign to the positive bias needed to explain the unexpected result. **PUB is NOT a viable candidate** for the omitted variable causing the unexpected positive price coefficient.

Candidate 2: INCt (average household income)

- **True effect of INC on car sales:** a new car is a normal good — higher household income should plausibly mean *more* new car sales, so the true coefficient of INC on SALES is expected to be **positive**.
- **Relationship between P (car price) and INC:** over time, rising average incomes and rising average new car prices (reflecting both general inflation and consumers trading up to higher-specification vehicles as they can afford to) would plausibly move together — suggesting a **positive** relationship between P and INC.
- **Expected bias** = (positive) × (positive) = **positive**.

Conclusion for INC: the expected bias from omitting INC is **positive** — exactly matching the direction of the unexpected result. **INC IS a viable candidate** for the variable whose omission caused the unexpected positive price coefficient: omitting household income let the genuine, positive income effect on car sales get wrongly attributed to price (since price and income are themselves positively correlated), biasing the estimated price coefficient upward from its true negative value into the observed positive one.

EXAM TECHNIQUE

This is a textbook example of the expected-bias-analysis technique: when a coefficient's sign contradicts theory, rather than guessing at possible omitted variables, systematically work through each candidate's own expected effect on the dependent variable and its expected relationship with the suspect included variable — only a candidate whose combined sign matches the actual anomaly is a real suspect.

Mark Allocation — Part (b) (30%)

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly states the expected bias formula (sign of omitted variable's own effect × sign of its relationship with the included variable) AND correctly works through PUB (negative bias, not viable) and INC (positive bias, viable) with clear economic reasoning for each of the two component signs in both cases.
Good	51-75%	Correct formula applied to both candidates, but the economic reasoning for one of the four component signs is weak or missing.
Partial	26-50%	Only one candidate correctly analysed.
Minimal	0-25%	No credible expected bias analysis given.

Question A2 Total: 100%

Section A · Question 3

Diagnosing autocorrelation and heteroscedasticity, the Durbin-Watson test, and testing with a lagged dependent variable — 100 marks

A3(a)

[30 marks]

A researcher runs a regression on a time-series dataset from 2001 to 2018 and gets a graph of actual and fitted values that cross repeatedly, with the actual series running above the fitted line for sustained multi-year periods and then below it for other sustained periods. (i) Is this regression likely to suffer from any econometric problem, and if so, which one? (ii) What is the likely cause of this econometric problem in this case? Explain your reasoning.

A3(b)

[30 marks]

A graph of residuals from a time-series regression shows the spread of the residuals widening steadily as the observation index increases, from a narrow band near zero at the start to a wide band (roughly -50 to $+50$) by the end. (i) What does this graph of residuals suggest? Explain your answer. (ii) Explain, with the aid of an example, why this might occur.

A3(c)

[20 marks]

For each of the following scenarios, state whether autocorrelation is present, based on the Durbin Watson test with a 5% level of significance. Briefly explain your answer. (i) $d=1.27$, $n=20$, $k=3$. (ii) $d=3.24$, $n=30$, $k=4$. (iii) $d=2.22$, $n=22$, $k=2$. (iv) $d=1.87$, $n=35$, $k=3$.

A3(d)

[20 marks]

A researcher proposes $FP_t = \beta_0 + \beta_1 FP_{t-1} + \beta_2 TS_t + \beta_3 MS_t + \beta_4 MA_t + \varepsilon_t$ for a football club's league finishing position over time, and believes it may suffer from autocorrelation. How would you recommend the researcher goes about testing this formally? Justify your answer.

FULL MODEL ANSWER

(i) The likely econometric problem: (positive first-order)

autocorrelation / serial correlation. The pattern described — actual values running consistently above the fitted line for sustained multi-year stretches, then consistently below it for other sustained stretches, rather than crossing the fitted line unpredictably and randomly — is the classic visual signature of positive autocorrelation: successive error terms are positively correlated with each other, so once the actual series drifts above (or below) the fitted trend, it tends to *stay* there for multiple consecutive periods rather than reverting immediately.

Actual vs. Fitted: The Signature Pattern of Positive Autocorrelation

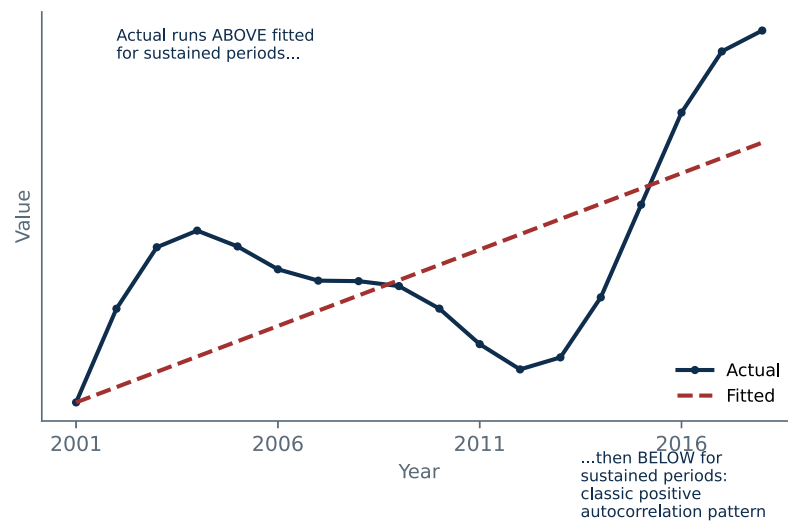


Fig. 1 — Illustrative actual-vs-fitted series showing the sustained-runs pattern (actual persistently above, then persistently below, the fitted line) that signals positive autocorrelation.

(ii) The likely cause: for annual, economically-themed time-series data like this, the most common underlying cause of this pattern is a **specification problem** — most often an **omitted variable that is itself trending or cyclical** (e.g. a genuine business-cycle or momentum effect in the dependent variable that isn't captured by the included regressors), or an **incorrect functional form** (e.g. fitting a straight line to a relationship that is genuinely non-linear over the sample period). Either specification error leaves a systematic, persistent pattern in the residuals that a well-specified model would have absorbed — this is sometimes called "impure" autocorrelation, as distinct from "pure" autocorrelation arising purely from genuine inertia in the underlying economic process itself. Since the specific cause can't be determined from the graph alone, the researcher's first step should be to carefully review the model's specification for a plausible omitted variable or functional-form issue before assuming the autocorrelation is a pure, unavoidable feature of the data.

Mark Allocation — Part (a) (30%)

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly identifies (positive) autocorrelation from the sustained-runs pattern described AND correctly identifies an omitted trending/cyclical variable or incorrect functional form as the likely cause, with the "impure vs. pure" distinction and the recommendation to check specification first.
Good	51-75%	Correctly identifies autocorrelation with a plausible cause given, but without the impure/pure distinction.
Partial	26-50%	Identifies "a pattern" in the residuals without correctly naming autocorrelation specifically.
Minimal	0-25%	No credible diagnosis given.

(i) What the graph suggests: heteroscedasticity — a non-constant error variance. The description given (a narrow band of residuals near zero early in the series, widening steadily to a much wider band, roughly -50 to $+50$, later on) is the classic "fan" or "megaphone" shape that is the standard visual diagnostic for heteroscedasticity: the spread (variance) of the residuals is clearly not constant across observations, violating the classical OLS assumption of homoscedasticity (constant error variance).

Residual Plot Showing Heteroscedasticity

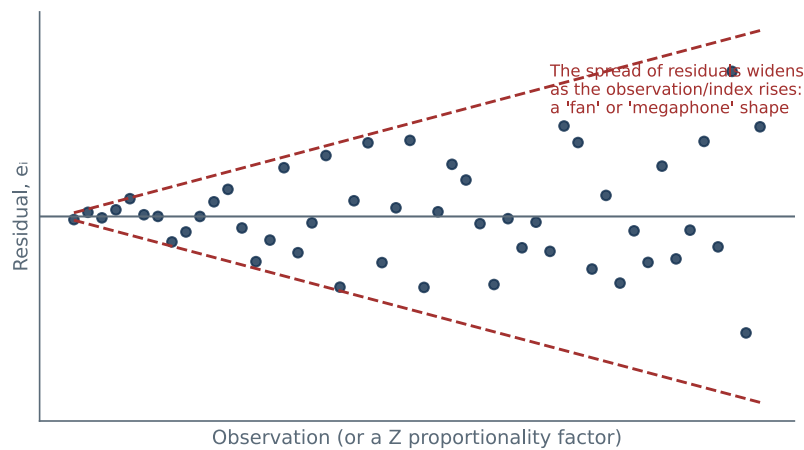


Fig. 2 — The classic 'fan' shape: residual spread widens steadily as the observation index increases, the standard visual diagnostic for heteroscedasticity.

(ii) Why this might occur, with an example: heteroscedasticity commonly arises when the **scale** of the phenomenon being modelled itself grows or varies substantially across the sample. A classic textbook example: regressing **household spending on household income** across households of widely differing income levels — low-income households have little discretion in how much they spend (most of a small income goes to necessities, leaving little room for variation), so their spending is tightly clustered around what the regression line predicts; high-income households, by contrast, have far more discretion over how much of their (much larger) income to spend versus save, so their actual spending varies much more widely around the predicted value. In a *time-series* context specifically (as here), the same underlying mechanism can appear if the scale of the dependent variable itself grows substantially over the sample period — for example, a national economic variable measured in absolute currency terms will naturally show larger absolute swings (and so larger absolute residuals) in later years, simply because the whole economy is bigger by then, even if the underlying relationship's *relative* volatility hasn't changed at all.

Mark Allocation — Part (b) (30%)

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly identifies heteroscedasticity from the widening-spread "fan" pattern AND gives a correct, well-explained example (household spending/income or an equivalent scale-driven case) showing genuine understanding of why variance grows with scale.
Good	51-75%	Correctly identifies heteroscedasticity with a less fully-developed example.
Partial	26-50%	Identifies "non-constant variance" without naming heteroscedasticity or giving a credible example.
Minimal	0-25%	No credible diagnosis given.

The Durbin-Watson Decision Rule

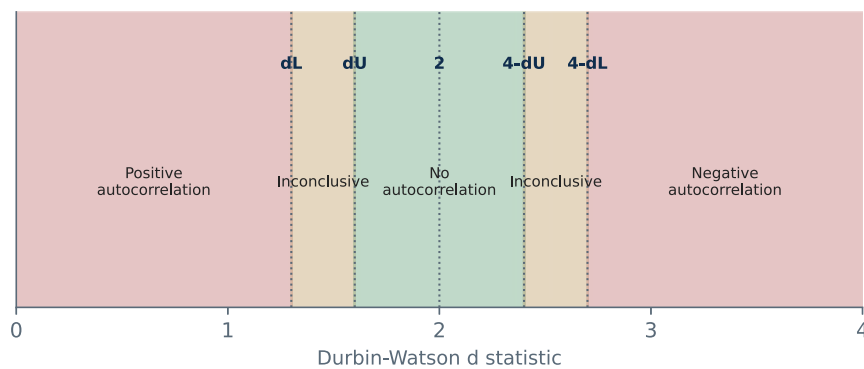


Fig. 3 — The five-zone Durbin-Watson decision rule: reject (positive autocorrelation) below d_L , inconclusive between d_L and d_U , fail to reject (no autocorrelation) between d_U and $4-d_U$, inconclusive between $4-d_U$ and $4-d_L$, and reject (negative autocorrelation) above $4-d_L$.

Scenario	d_L, d_U (from Appendix A5)	$4-d_U, 4-d_L$	Conclusion
(i) $d=1.27$, $n=20$, $k=3$	$d_L=1.100$, $d_U=1.537$	—	$d_L < d < d_U$ ($1.100 < 1.27 < 1.537$): INCONCLUSIVE (positive- autocorrelation zone)
(ii) $d=3.24$, $n=30$, $k=4$	$d_L=1.214$, $d_U=1.650$	$4-d_U=2.350$, $4-d_L=2.786$	$d > 4-d_L$ ($3.24 > 2.786$): NEGATIVE autocorrelation present
(iii) $d=2.22$, $n=22$, $k=2$	$d_L=1.239$, $d_U=1.429$	$4-d_U=2.571$, $4-d_L=2.761$	$d_U < d < 4-d_U$ ($1.429 < 2.22 < 2.571$): NO evidence of autocorrelation
(iv) $d=1.87$, $n=35$, $k=3$	$d_L=1.343$, $d_U=1.584$	$4-d_U=2.416$, $4-d_L=2.657$	$d_U < d < 4-d_U$ ($1.584 < 1.87 < 2.416$): NO evidence of autocorrelation

Mark Allocation — Part (c) (20%)

Tier	Marks	What separates this tier
Full credit	76-100%	All four scenarios correctly read from the Durbin-Watson table with the correct n/k row AND correctly placed into the correct zone of the five-zone decision rule, including correctly checking the negative-autocorrelation boundary ($4-d_U, 4-d_L$) for scenario (ii) specifically.
Good	51-75%	At least three of the four scenarios correctly decided.
Partial	26-50%	At least two of the four correctly decided, or all four attempted but only checking the positive-autocorrelation boundary (missing the negative-autocorrelation zone check for scenario (ii)).
Minimal	0-25%	No credible Durbin-Watson decisions given.

Why the standard Durbin-Watson test cannot be used here: the model as specified includes FP_{t-1} (**the lagged dependent variable**) as a **regressor** — the club's finishing position last year predicting this year's. This directly violates one of the three core assumptions required for the Durbin-Watson test to be valid: the regression model must **not** include a lagged dependent variable as an independent variable. Using the standard DW test

here would produce a statistic that is **biased toward 2** (the "no autocorrelation" value) regardless of whether genuine autocorrelation is present, making it systematically unreliable and likely to understate any true autocorrelation problem in exactly this kind of dynamic specification.

The recommended alternative — the Breusch-Godfrey (BG) test: the Breusch-Godfrey test is specifically designed to remain valid in cases the Durbin-Watson test cannot handle, including models with a **lagged dependent variable** as a regressor, **higher-order** autoregressive structures (AR(2), AR(3), etc., not just first-order AR(1)), and moving-average error terms.

- **Step 1:** estimate the original model (as specified, including FP_{t-1}) by OLS and obtain the residuals, e_t .
- **Step 2:** regress e_t on all the original regressors (FP_{t-1} , TS_t , MS_t , MA_t) *plus* p lagged values of the residuals themselves (e.g. e_{t-1} for a first-order test), and obtain the R^2 from this auxiliary regression.
- **Step 3a:** in large samples, $(n-p)R^2$ from the auxiliary regression follows a chi-square distribution with p degrees of freedom under the null hypothesis of no serial correlation — compare to the critical chi-square value.
- **Step 3b (alternative):** equivalently, the F-statistic from the auxiliary regression can be used, with $(p, n-k-p)$ degrees of freedom, where k is the number of parameters (including the intercept) in the auxiliary regression.
- **Conclusion:** rejecting the null hypothesis of no serial correlation ($\rho_1=\rho_2=\dots=\rho_p=0$) implies evidence of autocorrelation, exactly as with the Durbin-Watson test — but obtained via a method that remains valid given the lagged dependent variable in the original specification.

EXAM TECHNIQUE

Durbin's h-test is a narrower, classic alternative built for precisely this single case (a first-order AR(1) test with one lagged dependent variable), and is also creditable — but the Breusch-Godfrey test is the more general, and more commonly recommended, modern solution, since it additionally handles higher-order autocorrelation the researcher's model might also be exposed to given the annual sports-performance context (e.g. a club's underlying form persisting across more than just one lag).

Mark Allocation — Part (d) (20%)

Tier	Marks	What separates this tier
Full credit	76-100%	Correctly identifies that the lagged dependent variable (FP_{t-1}) invalidates the standard Durbin-Watson test, citing the specific assumption violated AND correctly recommends the Breusch-Godfrey test (or Durbin's h-test) as the valid alternative AND correctly outlines the BG test's steps (auxiliary regression on residuals plus lagged residuals, chi-square or F test).
Good	51-75%	Correctly identifies the DW test's invalidity and recommends Breusch-Godfrey, without the full auxiliary-regression steps.
Partial	26-50%	Recommends a formal autocorrelation test without correctly identifying why the standard Durbin-Watson test specifically cannot be used here.
Minimal	0-25%	No credible recommendation given.

Question A3 Total: 100%

